CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Mirror Maze

Goal: Arrange reflective surfaces to direct a light path through checkpoints to a target.

35 minutes | 8 steps | Engineering and Design | Playbook B meeting 33, on-track choice

Ask | Plan | Build | Test | Improve

You are done when

Trace the successful light path and explain one mirror-angle revision.

You will use

- Plastic mirrors
- Low-power classroom lights
- Target cards
- Grid paper
- Mirror stands
- Pencils

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place the light at the fixed source mark.
- Keep the power off.

2 Step 2: Light safety — 4 min

- Keep the light inside the table zone.
- Point no light toward a face.
- Point no reflection toward a window.
- Turn off the light before moving it.

3 Step 3: Predict the path — 4 min

- Mark the source.
- Mark each checkpoint.
- Draw the expected light path.

4 Step 4: Place the mirrors — 8 min

- Set one plastic mirror in its stand.
- Match its angle to the drawing.
- Repeat for each mirror.

5 Step 5: Trace the light — 7 min

- Turn on the fixed light.
- Trace the actual path on the grid.
- Mark the first missed checkpoint.

6 Step 6: Adjust the angles — 4 min

- Turn off the light.
- Adjust one mirror.
- Retest the full path.

7 Step 7: Power down — 2 min

- Turn off the light.
- Return mirrors to the padded tray.

8 Step 8: Exit check — 3 min

- Draw the final path.
- Circle the changed mirror.

What to turn in

The light path traced to the target and one angle fix.

No computer? Another way to do it

Trace paths with a ruler on paper using angle cards.

Finished early? Next challenge

One Less Mirror: Reach the same target with fewer reflective surfaces.

Stay safe

Never shine any light toward eyes, reflective clothing, windows, or outside the marked table zone.

Grade targets

Grade 5: Hit the target with two mirrors.
Grade 6: Use one less mirror and explain the angles.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.